Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2022	ENGI3491-WE01

Title:

Digital Electronics and Digital Signal Processing 3

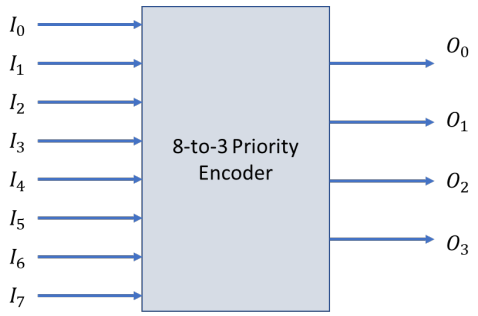
Time Allowed:	2 hours
Additional Material provided:	No
Reporting problems during the exam:	Queries about the instructions for this exam (rubric) should be directed to: eng.boe@durham.ac.uk. If you experience any difficulties uploading your exam answers to Gradescope report these immediately to: eng.boe@durham.ac.uk.

Instructions to Candidates:	Answer ALL questions. Please answer each question on a new page. All workings must be shown. This is an online open-book exam, submission is time-limited to 4 hours within a 24 hrs window. • *The examination attempt must be all your own work. You must not interact with another person (either one-to-one, by telephone, on the internet, in chat rooms, on social media, via smart watches, etc.), or use cheat sites, etc. whilst completing the exam. You must not discuss any examination papers with friends, relatives, current or past students of this or any other University before the end of the relevant 24-hour window.* • *Candidates are expected to answer all questions in their own words. Should close paraphrasing, direct quotation be necessary candidates must acknowledge the source of any material used in their answers (this includes text, images, diagrams, charts, tables and/or graphs). Full referencing is not expected, but a brief indication of the source of the material (e.g. name of author) must be given.*

Revision:	

Question 1

a) Figure Q1 shows the diagram and truth table of an 8-to-3 Priority Encoder. Using a PAL with 12 inputs and four outputs, implement an 8-to-3 Priority Encoder. Assume that each output OR gate of the PAL may have up to four AND inputs.



I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7	O_0	O_1	O_2	O_3
0	0	0	0	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	1
X	1	0	0	0	0	0	0	0	0	1	1
X	X	1	0	0	0	0	0	0	1	0	1
X	X	X	1	0	0	0	0	0	1	1	1
X	X	X	X	1	0	0	0	1	0	0	1
X	X	X	X	X	1	0	0	1	0	1	1
X	X	X	X	X	X	1	0	1	1	0	1
X	X	X	X	X	X	X	1	1	1	1	1

Figure Q1

[50%]

b) Write a VHDL code for a counter with three inputs: ‘T1’, clock signal ‘CLK’, and ‘Reset’, as well as one output ‘Out1’. ‘T1’ is an 8-bit input representing an 8-bit integer. The output, ‘Out1’ is a 4 bit output and counts up from 0 to 10 and back to 0, every T1 clock cycles. Use a state machine for your design and draw its diagram. Reset is an active-high asynchronous input that resets the output Out1 to ‘0’ when asserted. Use standard logic.

[50%]

continued

Question 2

a) i) Compare and contrast boundary scan testing and manual fault finding via circuit probing. Explain the roles of test vectors and the JTAG interface.

[40%]

b) An arbiter circuit is required to manage access to a shared resource for three asynchronous devices. The arbiter will have three 'Request' inputs, R, and three 'Grant access', G, outputs When a particular device needs access to the shared resource it passes a logic high to the arbiter via its request line and, if access is granted, the device will receive a logic high via its grant access line.

i) Explain why arbiters are used when sharing resources in this way with asynchronous devices.

ii) Design an asynchronous sequential logic circuit for an arbiter which is capable of exhibiting the behaviour described above. Your solution should include a sketch of the architecture at a block level, a state diagram, relevant working and clearly stated assumptions. You do not need to produce a detailed circuit diagram but should include the Boolean equations that would be necessary to implement the circuit. Note that these equations do not need to be simplified.

iii) As an alternative solution, it is suggested that the devices and the arbiter are implemented synchronously, i.e. with a common clock signal. Describe any disadvantages that may be encountered with this approach in a practical implementation and what additional circuit components are likely to be required versus your design in part (ii).

[60%]

END